Simulating Climate Risk Migration: The Shadow Banking Spillover Effect

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Most traditional papers look at how standard commercial banks price climate risk. My novel angle is to simulate what happens after a traditional bank correctly prices climate risk. Recent empirical evidence from the Indian credit market shows that after a localized natural disaster (like a severe flood), traditional bank branches systematically raise interest rates for all borrowers to price in the new risk (Abhiman Das, Tanmoy Majilla, Sumit Saurav). However, this credit contraction creates a vacuum. Unregulated or lightly regulated shadow banks, specifically Non-Banking Financial Companies (NBFCs), rapidly step in, expanding their lending in flood-affected areas to absorb the risky borrowers that traditional banks rejected (Sankalp Mathur, Ritwiz Sarma).

My project simulates a local economy to prove a macroeconomic point: strict climate risk regulations on traditional banks do not eliminate systemic climate risk; they merely force it into the opaque shadow banking sector.

This simulation project is divided into the following steps:

- Phase 1: Generating a synthetic economy
- Phase 2: Simulating climate shock and bank reaction
- Phase 3: Simulating the Shadow Banking Arbitrage/ Shadow Banking spillover
- Phase 4: Visualization and Results

[2]:
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# Dependencies

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

[4]:
```
# Plotting style

plt.style.use('seaborn-v0_8-whitegrid')
sns.set_context("talk")
```

1 Phase 1: Generating a synthetic economy

Methodology: This phase initializes a synthetic universe of 10,000 corporate entities to represent a diverse local economy. We use normal and uniform statistical distributions to generate realistic

baseline financial profiles for each firm. We then establish each firm's baseline credit health before any climate events occur.

Key Variables:

- ***revenue*** & ***ebitda_margin***: Simulated financial size and profitability margins for each firm.
- ***total_debt***: The firm's leverage, calculated as a multiple of revenue.
- ***flood_vulnerability***: A randomized score from 1 to 100 representing the firm's physical climate risk exposure.
- ***baseline_rate***: The standard interest rate (8.5%) charged by traditional banks in a normal macroeconomic environment.
- ***baseline_icr***: The Interest Coverage Ratio ($ICR = EBITDA/InterestExpense$). This is our primary metric for financial health. The firm's **Interest Coverage Ratio (ICR)** measures how many times its EBITDA covers its interest expense.

We simulate our economy such that it is consistent with the RBI's reports on Interest Coverage Ratio of Indian firms.

Based on the Reserve Bank of India's (RBI) Reports, the Interest Coverage Ratio (ICR) for Indian firms is generally assessed across different risk thresholds rather than as a single aggregate average, as it varies significantly depending on the sector and size of the firm.

RBI data indicates the following general baseline distribution for listed private non-financial companies in India:

- **Strong Baseline (ICR $>$ 4):** Roughly half of the companies evaluated (between 47% and 52% in recent years) operate with a very comfortable baseline ICR of greater than 4.
- **Moderate Baseline (1 $<$ ICR 4):** Approximately 25% to 30% of companies maintain an ICR within this middle tier.
- **Vulnerable Baseline (ICR 1):** Around 20% to 35% of companies historically operate with an ICR of 1 or below, meaning their operating profits are insufficient to cover their current debt interest expenses.

As a general financial benchmark, an ICR falling below 1.5 is widely considered a sign of acute financial stress for a company, signaling that the firm has a heavy debt burden and a high probability of defaulting on its interest obligations.

Source: RBI's report dated 28 Dec, 2023

Table 1.3: Debt Share of Firms Below/ Above ICR Threshold Values

Items	2016-17	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	H1: 2023-24
No of Companies	3,238	3,231	3,205	3,163	3,103	2,963	2,990	2,880
ICR <=1 (per cent)	35.5	34.2	34.1	37.6	34.7	32.6	21.0	21.9

Items	2016-17	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	H1: 2023-24
1 < ICR <= 4 (per cent)	34.8	34.3	27.6	25.2	31.5	15.2	27.9	30.5
ICR > 4 (per cent)	29.6	31.5	38.3	37.2	33.8	52.2	51.2	47.6

Sources: Capitaline and RBI staff calculations.

How to Interpret the Interest Coverage Ratio (ICR)

ICR	Interpretation	Meaning
ICR < 1	**Insufficient Coverage**	EBITDA is lower than interest expense. The firm cannot cover all its interest costs from operating earnings alone.
ICR 1–2	**Limited Buffer**	Interest is covered, but a decline in earnings or increase in borrowing costs could create financial pressure.
ICR > 2	**Larger Coverage Buffer**	EBITDA is more than twice interest expense, providing more room to absorb financial shocks, all else equal.

[5]:
```
print(" PHASE 1: GENERATING A SYNTHETIC ECONOMY ")

import pandas as pd
import numpy as np

np.random.seed(101)
n_firms = 10000

# 1.1 Generate Base Financials for 10,000 Firms

# Revenue in millions of Rupees, normal distribution
revenue = np.random.normal(loc=500, scale=150, size=n_firms)
revenue = np.clip(revenue, 100, 2000) # Ensures no negative revenues

# EBITDA Margin (Operating Profit Margin) between 10% and 30%
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ebitda_margin = np.random.uniform(0.10, 0.30, size=n_firms)
ebitda = revenue * ebitda_margin

# Total Debt (Leverage) - Segmented to match RBI ICR distribution targets
u = np.random.rand(n_firms)
debt_to_revenue = np.zeros(n_firms)

# ~50% Strong (ICR > 4) -> Assigned lower debt multiples
mask_strong = u < 0.50
debt_to_revenue[mask_strong] = np.random.uniform(0.1, 0.6, size=mask_strong.
 ↪sum())

# ~25% Moderate (1 < ICR <= 4) -> Assigned medium debt multiples
mask_moderate = (u >= 0.50) & (u <= 0.75)
debt_to_revenue[mask_moderate] = np.random.uniform(0.6, 2.5, size=mask_moderate.
 ↪sum())

# ~25% Vulnerable (ICR <= 1) -> Assigned higher debt multiples
mask_vulnerable = u > 0.75
debt_to_revenue[mask_vulnerable] = np.random.uniform(2.5, 4.5,␣
 ↪size=mask_vulnerable.sum())

total_debt = revenue * debt_to_revenue

# 1.2 Assign Climate Risk (Flood Vulnerability Score 1-100)

flood_vulnerability = np.random.randint(1, 101, size=n_firms)

# 1.3 Baseline Banking Profile (Pre-Shock)

baseline_rate = 0.085 # 8.5% average traditional bank loan rate
interest_expense = total_debt * baseline_rate
baseline_icr = ebitda / interest_expense # Interest Coverage Ratio

# DataFrame

df = pd.DataFrame({
    'Firm_ID': range(1, n_firms + 1),
    'Revenue': revenue,
    'EBITDA': ebitda,
    'Total_Debt': total_debt,
    'Flood_Vulnerability': flood_vulnerability,
    'Baseline_Interest_Expense': interest_expense,
    'Baseline_ICR': baseline_icr
})

print(f"Generated financial profiles for {n_firms} synthetic firms.")
```

```
print(f"Strong Baseline (ICR > 4): {(df['Baseline_ICR'] > 4).sum()} firms␣
  ↪({(df['Baseline_ICR'] > 4).sum()/n_firms*100:.1f}%)")
print(f"Moderate Baseline (1 < ICR <= 4): {((df['Baseline_ICR'] > 1) &␣
  ↪(df['Baseline_ICR'] <= 4)).sum()} firms ({((df['Baseline_ICR'] > 1) &␣
  ↪(df['Baseline_ICR'] <= 4)).sum()/n_firms*100:.1f}%)")
print(f"Vulnerable Baseline (ICR <= 1): {(df['Baseline_ICR'] <= 1).sum()} firms␣
  ↪({(df['Baseline_ICR'] <= 1).sum()/n_firms*100:.1f}%)\n")
```

```
 PHASE 1: GENERATING A SYNTHETIC ECONOMY
Generated financial profiles for 10000 synthetic firms.
Strong Baseline (ICR > 4): 4380 firms (43.8%)
Moderate Baseline (1 < ICR <= 4): 2899 firms (29.0%)
Vulnerable Baseline (ICR <= 1): 2721 firms (27.2%)
```

[6]:
```
# Simulated dataset after Phase 1
df
```

[6]:

	Firm_ID	Revenue	EBITDA	Total_Debt	Flood_Vulnerability	Baseline_Interest_Expense	Baseline_ICR
0	1	906.027476	144.250235	1137.996096	2	96.729668	1.491272
1	2	594.219906	130.944611	300.868018	43	25.573782	5.120268
2	3	636.195417	90.459613	102.321418	62	8.697321	10.400860
3	4	575.573863	131.570884	1253.988631	2	106.589034	1.234375
4	5	597.667692	171.915680	171.452178	96	14.573435	11.796510
…	…	…	…	…	…	…	…
9995	9996	493.114572	59.423721	984.465739	30	83.679588	0.710134
9996	9997	507.803586	65.578099	236.042205	97	20.063587	3.268513
9997	9998	620.447869	114.976870	833.735111	77	70.867484	1.622421
9998	9999	651.604744	186.605469	1724.742473	20	146.603110	1.272862
9999	10000	519.685779	73.982430	167.755386	18	14.259208	5.188397

[10000 rows x 7 columns]

[7]:
```
# Summary statistics
df.describe()
```

[7]:

	Firm_ID	Revenue	EBITDA	Total_Debt	\
count	10000.00000	10000.000000	10000.000000	10000.000000	
mean	5000.50000	502.375788	100.826358	717.894951	
std	2886.89568	151.334776	42.610649	735.199878	
min	1.00000	100.000000	10.806847	11.092063	
25%	2500.75000	398.669332	68.683534	162.706774	
50%	5000.50000	501.358201	95.039859	336.095622	
75%	7500.25000	604.402655	128.453892	1168.661435	
max	10000.00000	1139.093225	296.345709	4484.928590	

	Flood_Vulnerability	Baseline_Interest_Expense	Baseline_ICR
count	10000.000000	10000.000000	10000.000000
mean	50.510600	61.021071	4.862628
std	28.552906	62.491990	5.294833
min	1.000000	0.942825	0.265337
25%	26.000000	13.830076	0.928975
50%	51.000000	28.568128	3.118729
75%	75.000000	99.336222	6.705314
max	100.000000	381.218930	33.356524

The financial variables and flood vulnerability scores are generated independently. As a result, flood vulnerability does not directly affect baseline financial health.

2 Phase 2: Simulating climate shock and bank reaction

Methodology: This phase introduces a physical climate shock (a severe flood) and models the subsequent behavioral reaction of traditional commercial banks. The model directly links the firm's vulnerability score to its revenue destruction. We then simulate traditional banks de-risking their portfolios by hiking interest rates for vulnerable borrowers and strictly cutting off credit to firms that fall below a safe regulatory health threshold.

Key Variables:

- ***shock_severity***: A percentage drop in revenue scaled directly to the firm's ***flood_vulnerability*** score (up to a 35% drop).
- ***Bank_Rate_Post_Shock***: The new interest rate. Banks hike this by 50 basis points for firms with a vulnerability score > 50.
- ***Stressed_Bank_ICR***: The recalculated Interest Coverage Ratio incorporating both the revenue shock and the higher interest burden.
- ***Bank_Approved***: A boolean flag. Traditional banks reject the loan renewal if the firm's stressed ICR falls below 1.5.

[9]:
```
print(" PHASE 2: SIMULATING CLIMATE SHOCK (FLOOD) ")

# 2.1 The Physical Shock: Flood severely impacts high-vulnerability firms
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# Revenue drops up to 35% for the most vulnerable firms

shock_severity = (df['Flood_Vulnerability'] / 100) * 0.35
df['Shocked_Revenue'] = df['Revenue'] * (1 - shock_severity)
df['Shocked_EBITDA'] = df['Shocked_Revenue'] * ebitda_margin

# 2.2 Traditional Bank Reaction: De-risking

# Banks hike rates by 50 basis points (0.005) for vulnerable firms (Score > 50)
df['Bank_Rate_Post_Shock'] = np.where(df['Flood_Vulnerability'] > 50,␣
 ↪baseline_rate + 0.005, baseline_rate)
df['Bank_Interest_Post_Shock'] = df['Total_Debt'] * df['Bank_Rate_Post_Shock']

# 2.3 Recalculate Health (Stressed ICR)

df['Stressed_Bank_ICR'] = df['Shocked_EBITDA'] / df['Bank_Interest_Post_Shock']

# 2.4 Bank Decision: Deny renewal if Stressed ICR < 1.5 (Strict regulatory␣
 ↪cutoff)

df['Bank_Approved'] = df['Stressed_Bank_ICR'] >= 1.5
rejected_count = (~df['Bank_Approved']).sum()

print(f"Climate shock applied. Traditional banks raised rates by 50 bps for␣
 ↪vulnerable firms.")
print(f"Traditional banks rejected {rejected_count} firms due to deteriorating␣
 ↪credit health.\n")
```

```
 PHASE 2: SIMULATING CLIMATE SHOCK (FLOOD)
Climate shock applied. Traditional banks raised rates by 50 bps for vulnerable
firms.
Traditional banks rejected 4104 firms due to deteriorating credit health.
```

[10]:
```
# Simulated dataset after Phase 2
df
```

[10]:

	Firm_ID	Revenue	EBITDA	Total_Debt	Flood_Vulnerability	\
0	1	906.027476	144.250235	1137.996096	2	
1	2	594.219906	130.944611	300.868018	43	
2	3	636.195417	90.459613	102.321418	62	
3	4	575.573863	131.570884	1253.988631	2	
4	5	597.667692	171.915680	171.452178	96	
…	…	…	…	…	…	
9995	9996	493.114572	59.423721	984.465739	30	
9996	9997	507.803586	65.578099	236.042205	97	

```
9997     9998  620.447869  114.976870   833.735111                   77
9998     9999  651.604744  186.605469  1724.742473                   20
9999    10000  519.685779   73.982430   167.755386                   18

      Baseline_Interest_Expense  Baseline_ICR  Shocked_Revenue  \
0                        96.729668      1.491272       899.685284
1                        25.573782      5.120268       504.789810
2                         8.697321     10.400860       498.141011
3                       106.589034      1.234375       571.544846
4                        14.573435     11.796510       396.851348
…                              …             …              …
9995                     83.679588      0.710134       441.337542
9996                     20.063587      3.268513       335.404268
9997                     70.867484      1.622421       453.237168
9998                    146.603110      1.272862       605.992412
9999                     14.259208      5.188397       486.945575

      Shocked_EBITDA  Bank_Rate_Post_Shock  Bank_Interest_Post_Shock  \
0         143.240483                 0.085                  96.729668
1         111.237447                 0.085                  25.573782
2          70.829877                 0.090                   9.208928
3         130.649888                 0.085                 106.589034
4         114.152012                 0.090                  15.430696
…                …                     …                         …
9995       53.184230                 0.085                  83.679588
9996       43.314335                 0.090                  21.243798
9997       83.990604                 0.090                  75.036160
9998      173.543087                 0.085                 146.603110
9999       69.321537                 0.085                  14.259208

      Stressed_Bank_ICR  Bank_Approved
0              1.480833          False
1              4.349668           True
2              7.691436           True
3              1.225735          False
4              7.397723           True
…                   …              …
9995           0.635570          False
9996           2.038917           True
9997           1.119335          False
9998           1.183761          False
9999           4.861528           True

[10000 rows x 13 columns]
```

[15]:
```
# Summary statistics after the climate shock
df.describe()
```

[15]:

	Firm_ID	Revenue	EBITDA	Total_Debt
count	10000.00000	10000.000000	10000.000000	10000.000000
mean	5000.50000	502.375788	100.826358	717.894951
std	2886.89568	151.334776	42.610649	735.199878
min	1.00000	100.000000	10.806847	11.092063
25%	2500.75000	398.669332	68.683534	162.706774
50%	5000.50000	501.358201	95.039859	336.095622
75%	7500.25000	604.402655	128.453892	1168.661435
max	10000.00000	1139.093225	296.345709	4484.928590

	Flood_Vulnerability	Baseline_Interest_Expense	Baseline_ICR
count	10000.000000	10000.000000	10000.000000
mean	50.510600	61.021071	4.862628
std	28.552906	62.491990	5.294833
min	1.000000	0.942825	0.265337
25%	26.000000	13.830076	0.928975
50%	51.000000	28.568128	3.118729
75%	75.000000	99.336222	6.705314
max	100.000000	381.218930	33.356524

	Shocked_Revenue	Shocked_EBITDA	Bank_Rate_Post_Shock
count	10000.000000	10000.000000	10000.000000
mean	413.753220	83.040800	0.087507
std	135.934318	36.820156	0.002500
min	65.350000	7.327042	0.085000
25%	318.986951	55.388995	0.085000
50%	406.990576	77.458444	0.090000
75%	501.814713	106.232702	0.090000
max	1013.188142	249.976707	0.090000

	Bank_Interest_Post_Shock	Stressed_Bank_ICR	Final_Interest_Expense
count	10000.000000	10000.000000	10000.000000
mean	62.819996	3.901649	84.658305
std	64.379232	4.336339	94.544374
min	0.942825	0.175438	0.942825
25%	14.214210	0.747016	14.240288
50%	29.376347	2.453253	29.973833
75%	101.896685	5.364617	145.707467
max	381.218930	31.187908	560.616074

	Final_ICR
count	10000.000000
mean	3.813584
std	4.399422
min	0.126315
25%	0.520459
50%	2.453253

```
75%         5.364617
max        31.187908
```

3 Phase 3: Shadow banking (NBFC) spillover

Methodology: This phase simulates the macroeconomic spillover effect. Firms that are cut off by traditional banks do not simply disappear; they are forced to seek funding in the unregulated shadow banking sector (Non-Banking Financial Companies, or NBFCs). The model assumes NBFCs will absorb these risky borrowers but will charge a massive "climate risk premium," leading to a surge in technical defaults.

Key Variables:

- ***nbfc_rate***: The predatory shadow banking interest rate, modeled as a 400 basis point (4.0%) premium over the baseline rate.
- ***Final_Lender***: Tracks whether the firm stayed with a 'Traditional Bank' or was pushed to a 'Shadow Bank'.
- ***Final_ICR***: The ultimate Interest Coverage Ratio after accounting for NBFC interest rates.
- ***Default***: A boolean flag triggered if a firm's `Final_ICR` drops below 1.0, indicating it no longer generates enough operating profit to service its shadow debt.

[18]:
```
print(" PHASE 3: SHADOW BANKING SPILLOVER ")

# 3.1 NBFCs absorb the rejected firms but charge a massive risk premium (400␣
 ↪bps / 4.0%)

nbfc_rate = baseline_rate + 0.040
df['Final_Lender'] = np.where(df['Bank_Approved'], 'Traditional Bank', 'Shadow␣
 ↪Bank (NBFC)')

# Calculating NBFC interest burden

df['Final_Interest_Expense'] = np.where(
    df['Bank_Approved'],
    df['Bank_Interest_Post_Shock'],
    df['Total_Debt'] * nbfc_rate
)

# 3.2 Final Firm Health and Default Calculation

df['Final_ICR'] = df['Shocked_EBITDA'] / df['Final_Interest_Expense']

# A firm defaults if its final ICR drops below 1.0 (cannot cover interest␣
 ↪payments)

df['Default'] = df['Final_ICR'] < 1.0

nbfc_count = (df['Final_Lender'] == 'Shadow Bank (NBFC)').sum()
```

```
default_count = df['Default'].sum()

print(f"Shadow Banks absorbed {nbfc_count} rejected firms.")
print(f"Total Defaults caused by predatory shadow rates: {default_count} firms.
  ↪\n")
```

```
 PHASE 3: SHADOW BANKING SPILLOVER
Shadow Banks absorbed 4104 rejected firms.
Total Defaults caused by predatory shadow rates: 4013 firms.
```

[19]:
```
# Simulated dataset after Phase 3
df
```

[19]:

	Firm_ID	Revenue	EBITDA	Total_Debt	Flood_Vulnerability	\
0	1	906.027476	144.250235	1137.996096	2	
1	2	594.219906	130.944611	300.868018	43	
2	3	636.195417	90.459613	102.321418	62	
3	4	575.573863	131.570884	1253.988631	2	
4	5	597.667692	171.915680	171.452178	96	
…	…	…	…	…	…	
9995	9996	493.114572	59.423721	984.465739	30	
9996	9997	507.803586	65.578099	236.042205	97	
9997	9998	620.447869	114.976870	833.735111	77	
9998	9999	651.604744	186.605469	1724.742473	20	
9999	10000	519.685779	73.982430	167.755386	18	

	Baseline_Interest_Expense	Baseline_ICR	Shocked_Revenue	\
0	96.729668	1.491272	899.685284	
1	25.573782	5.120268	504.789810	
2	8.697321	10.400860	498.141011	
3	106.589034	1.234375	571.544846	
4	14.573435	11.796510	396.851348	
…	…	…	…	
9995	83.679588	0.710134	441.337542	
9996	20.063587	3.268513	335.404268	
9997	70.867484	1.622421	453.237168	
9998	146.603110	1.272862	605.992412	
9999	14.259208	5.188397	486.945575	

	Shocked_EBITDA	Bank_Rate_Post_Shock	Bank_Interest_Post_Shock	\
0	143.240483	0.085	96.729668	
1	111.237447	0.085	25.573782	
2	70.829877	0.090	9.208928	
3	130.649888	0.085	106.589034	
4	114.152012	0.090	15.430696	
…	…	…	…	
9995	53.184230	0.085	83.679588	

```
9996       43.314335                 0.090                 21.243798
9997       83.990604                 0.090                 75.036160
9998      173.543087                 0.085                146.603110
9999       69.321537                 0.085                 14.259208

      Stressed_Bank_ICR  Bank_Approved        Final_Lender  \
0              1.480833          False  Shadow Bank (NBFC)
1              4.349668           True    Traditional Bank
2              7.691436           True    Traditional Bank
3              1.225735          False  Shadow Bank (NBFC)
4              7.397723           True    Traditional Bank
…                 …              …                  …
9995           0.635570          False  Shadow Bank (NBFC)
9996           2.038917           True    Traditional Bank
9997           1.119335          False  Shadow Bank (NBFC)
9998           1.183761          False  Shadow Bank (NBFC)
9999           4.861528           True    Traditional Bank

      Final_Interest_Expense  Final_ICR  Default
0                 142.249512   1.006966    False
1                  25.573782   4.349668    False
2                   9.208928   7.691436    False
3                 156.748579   0.833500     True
4                  15.430696   7.397723    False
…                        …          …        …
9995              123.058217   0.432188     True
9996               21.243798   2.038917    False
9997              104.216889   0.805921     True
9998              215.592809   0.804958     True
9999               14.259208   4.861528    False

[10000 rows x 17 columns]
```

[14]:
```
# Summary statistics after Phase 3:
df.describe()
```

[14]:
```
           Firm_ID       Revenue        EBITDA    Total_Debt  \
count  10000.00000  10000.000000  10000.000000  10000.000000
mean    5000.50000    502.375788    100.826358    717.894951
std     2886.89568    151.334776     42.610649    735.199878
min        1.00000    100.000000     10.806847     11.092063
25%     2500.75000    398.669332     68.683534    162.706774
50%     5000.50000    501.358201     95.039859    336.095622
75%     7500.25000    604.402655    128.453892   1168.661435
max    10000.00000   1139.093225    296.345709   4484.928590

       Flood_Vulnerability  Baseline_Interest_Expense  Baseline_ICR  \
```

```
count         10000.000000               10000.000000  10000.000000
mean             50.510600                  61.021071      4.862628
std              28.552906                  62.491990      5.294833
min               1.000000                   0.942825      0.265337
25%              26.000000                  13.830076      0.928975
50%              51.000000                  28.568128      3.118729
75%              75.000000                  99.336222      6.705314
max             100.000000                 381.218930     33.356524

       Shocked_Revenue  Shocked_EBITDA  Bank_Rate_Post_Shock  \
count     10000.000000    10000.000000          10000.000000
mean        413.753220       83.040800              0.087507
std         135.934318       36.820156              0.002500
min          65.350000        7.327042              0.085000
25%         318.986951       55.388995              0.085000
50%         406.990576       77.458444              0.090000
75%         501.814713      106.232702              0.090000
max        1013.188142      249.976707              0.090000

       Bank_Interest_Post_Shock  Stressed_Bank_ICR  Final_Interest_Expense  \
count              10000.000000       10000.000000            10000.000000
mean                  62.819996           3.901649               84.658305
std                   64.379232           4.336339               94.544374
min                    0.942825           0.175438                0.942825
25%                   14.214210           0.747016               14.240288
50%                   29.376347           2.453253               29.973833
75%                  101.896685           5.364617              145.707467
max                  381.218930          31.187908              560.616074

          Final_ICR
count  10000.000000
mean       3.813584
std        4.399422
min        0.126315
25%        0.520459
50%        2.453253
75%        5.364617
max       31.187908
```

4 Phase 4: Visualization and Results

Methodology: This final phase aggregates the firm-level microdata into macroeconomic totals to prove our thesis. I use ***matplotlib*** and ***seaborn*** to generate three visual proofs: a portfolio migration bar chart, a shadow-bank default pie chart, and a density distribution of corporate interest burdens.

Key Outputs:

- **Migration of Loan Portfolios:** Visually demonstrates the credit vacuum, how traditional bank loan volume shrinks while shadow bank volume explodes.
- **Survival Rate in Shadow Banking:** Highlights the fatal consequences of the risk premium, showing the exact percentage of defaults within the NBFC sector.
- **Shift in Interest Expense Burden:** A Kernel Density Estimate (KDE) plot showing the systemic rightward shift in debt-servicing costs across the entire simulated economy.

4.1 Visualization

4.1.1 Plot 1: Migration of Loan Portfolios

[11]:
```
# Viridis palette
viridis_colors = sns.color_palette("viridis", 6)

# Bar chart
plt.figure(figsize=(8, 6))

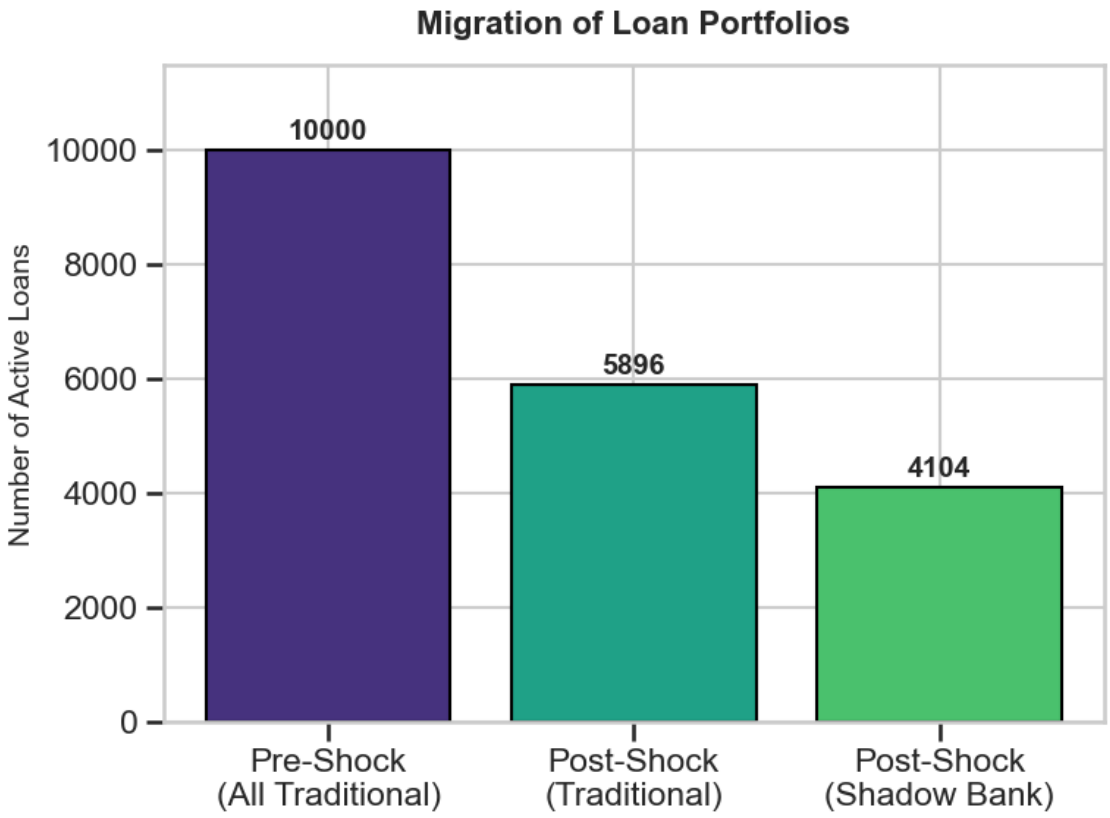
labels = ['Pre-Shock\n(All Traditional)', 'Post-Shock\n(Traditional)',␣
 ↪'Post-Shock\n(Shadow Bank)']
loan_counts = [n_firms, df['Bank_Approved'].sum(), nbfc_count]

bar_colors = [viridis_colors[0], viridis_colors[3], viridis_colors[4]]

plt.bar(labels, loan_counts, color=bar_colors, edgecolor='black', linewidth=1.5)
for i, v in enumerate(loan_counts):
    plt.text(i, v + 200, str(v), ha='center', fontweight='bold', fontsize=14)

plt.title('Migration of Loan Portfolios', fontsize=16, fontweight='bold',␣
 ↪pad=15)
plt.ylabel('Number of Active Loans', fontsize=14)
plt.ylim(0, n_firms + 1500)

plt.tight_layout()
plt.show()
```



4.1.2 Plot 2: Survival Rate in Shadow Banking

[12]:
```
plt.figure(figsize=(8, 6))

# Defaults specifically within the Shadow Banking portfolio

nbfc_defaults = df[(df['Final_Lender'] == 'Shadow Bank (NBFC)') &␣
 ↪(df['Default'] == True)].shape[0]
nbfc_survivals = nbfc_count - nbfc_defaults

# Calculate percentages

pct_default = (nbfc_defaults / nbfc_count) * 100
pct_survive = (nbfc_survivals / nbfc_count) * 100

pie_labels = [f'Defaulted\n(Shadow Debt)\n{pct_default:.1f}%',
              f'Survived\n(Shadow Debt)\n{pct_survive:.1f}%']
pie_sizes = [nbfc_defaults, nbfc_survivals]

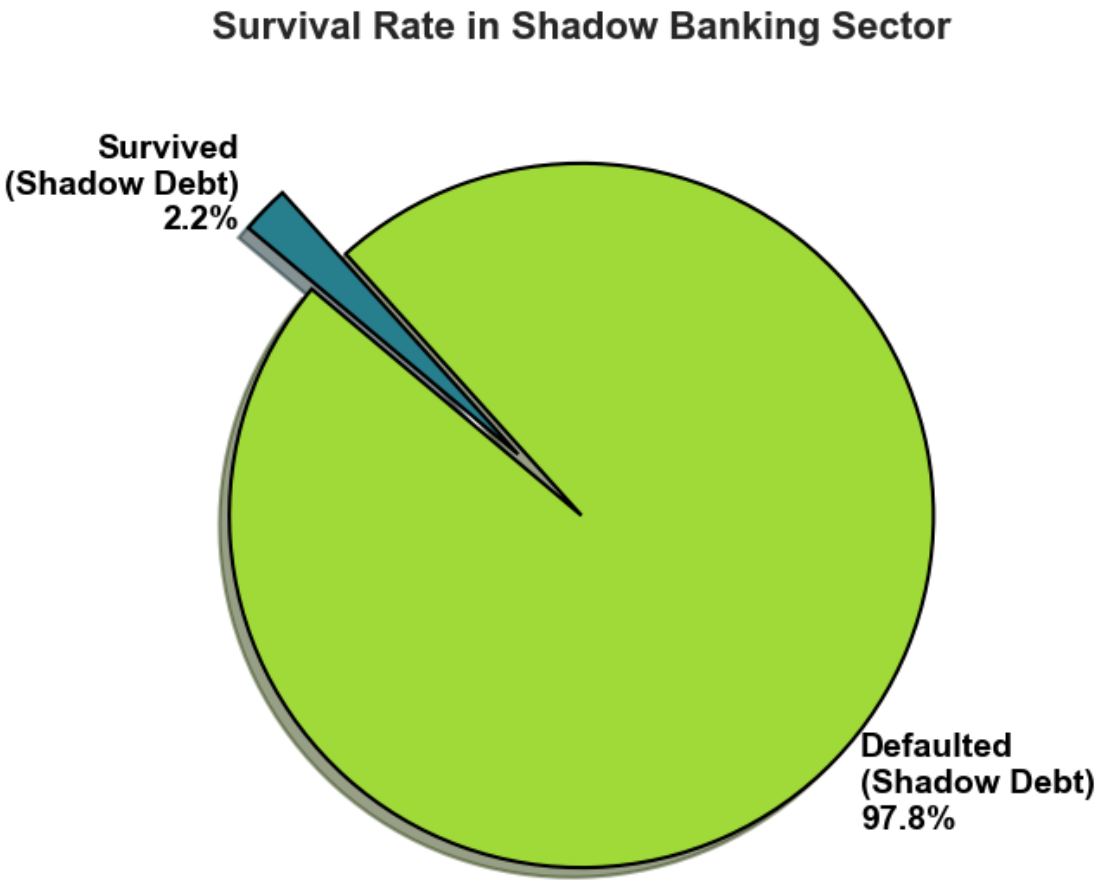
pie_colors = [viridis_colors[5], viridis_colors[2]]
```

```
explode = (0, 0.25)

plt.pie(pie_sizes, explode=explode, labels=pie_labels, colors=pie_colors,
            shadow=True, startangle=140,
            textprops={'fontsize': 14, 'weight': 'bold', 'color': 'black'},
            wedgeprops={'edgecolor': 'black', 'linewidth': 1.5})

plt.title('Survival Rate in Shadow Banking Sector', fontsize=16,␣
 ↪fontweight='bold', pad=15)

plt.tight_layout()
plt.show()
```



4.1.3 Plot 3: Shift in Interest Expense Burden

[13]:
```
plt.figure(figsize=(10, 6))

# KDE plot

sns.kdeplot(data=df, x='Baseline_Interest_Expense', color=viridis_colors[0],
            label='Pre-Shock Burden', fill=True, alpha=0.5, linewidth=2)
sns.kdeplot(data=df, x='Final_Interest_Expense', color=viridis_colors[4],
            label='Post-Shock Burden', fill=True, alpha=0.6, linewidth=2)

plt.title('Shift in Corporate Interest Expense Burden', fontsize=16,␣
 ↪fontweight='bold', pad=15)
plt.xlabel('Annual Interest Expense (Millions)', fontsize=14)
plt.ylabel('Density of Firms', fontsize=14)
plt.legend()

plt.tight_layout()
plt.show()
```



4.2 Results

[ ]:
```
print(f"1. I simulated a synthetic economy of {n_firms} firms to test␣
 ↪regulatory climate risk behavior.")
```

```
print(f"2. After a simulated severe flood shock, traditional banks aggressively␣
 ↪de-risked. They hiked interest rates by 50 basis points for highly␣
 ↪vulnerable firms and outright rejected {nbfc_count} firms whose Interest␣
 ↪Coverage Ratios dropped below 1.5.")
print(f"3. While the traditional banking sector looked safer on paper, the␣
 ↪macroeconomic risk did not disappear. It migrated.")
print(f"4. Unregulated Shadow Banks (NBFCs) stepped in to absorb all␣
 ↪{nbfc_count} rejected firms, but charged an exorbitant 400 basis point risk␣
 ↪premium.")
print(f"5. Because highly vulnerable firms were forced to take on extremely␣
 ↪expensive shadow debt just to survive the shock, {nbfc_defaults} of them␣
 ↪collapsed into default.")
print("6. CONCLUSION: Traditional bank climate stress tests are fundamentally␣
 ↪flawed if they ignore shadow banking. Pushing climate risk out of regulated␣
 ↪banks actually increases total systemic defaults.")
```

1. I simulated a synthetic economy of 10000 firms to test regulatory climate risk behavior.
2. After a simulated severe flood shock, traditional banks aggressively de-risked. They hiked interest rates by 50 basis points for highly vulnerable firms and outright rejected 4104 firms whose Interest Coverage Ratios dropped below 1.5.
3. While the traditional banking sector looked safer on paper, the macroeconomic risk did not disappear. It migrated.
4. Unregulated Shadow Banks (NBFCs) stepped in to absorb all 4104 rejected firms, but charged an exorbitant 400 basis point risk premium.
5. Because highly vulnerable firms were forced to take on extremely expensive shadow debt just to survive the shock, 4013 of them collapsed into default.
6. ***CONCLUSION***: Traditional bank climate stress tests are fundamentally flawed if they ignore shadow banking. Pushing climate risk out of regulated banks actually increases total systemic defaults.